



Lakshya JEE 2.0(2024)

Indefinite Integration

DPP-03

1. $\int \frac{dx}{\sqrt{7-x^2}} =$

(1) $-\sin^{-1}\left(\frac{x}{\sqrt{7}}\right) + c$

(2) $\cos^{-1}\left(\frac{x}{\sqrt{7}}\right) + c$

(3) $-\cos^{-1}\left(\frac{x}{\sqrt{7}}\right) + c$

(4) $\sin^{-1}\left(\frac{x}{\sqrt{7}}\right) + c$

2. $\int \frac{e^x dx}{\sqrt{e^{2x}-1}}$

(1) $\ln\left(e^x + \sqrt{e^{2x}-1}\right) + c$

(2) $-\ln\left(e^x + \sqrt{e^{2x}-1}\right) + c$

(3) $\ln\left(e^x - \sqrt{e^{2x}-1}\right) + c$

(4) $\ln\left(e^x + \sqrt{e^{2x}+1}\right) + c$

3. $\int \frac{e^x dx}{4+e^{2x}}$

(1) $-\frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right) + c$

(2) $\frac{1}{2}\tan^{-1}\left(\frac{e^x}{4}\right) + c$

(3) $\frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right) + c$

(4) $\frac{1}{2}\tan^{-1}\left(\frac{e^x}{6}\right) + c$

4. $\int \frac{dx}{4x^2+9}$

(1) $\frac{1}{4}\tan^{-1}\left(\frac{2x}{3}\right) + c$

(2) $\frac{1}{8}\tan^{-1}\left(\frac{2x}{3}\right) + c$

(3) $\frac{1}{12}\tan^{-1}\left(\frac{2x}{3}\right) + c$

(4) $\frac{1}{6}\tan^{-1}\left(\frac{2x}{3}\right) + c$

5. $\int \frac{dx}{\cos x - \sin x} = \frac{1}{\sqrt{2}} \log|f(x)| + c$, then $f(x)$ equals:

(1) $\tan \frac{x}{2}$

(2) $\cot \frac{x}{2}$

(3) $\tan\left(\frac{x}{2} + \frac{3\pi}{8}\right)$

(4) $\tan\left(\frac{x}{2} + \frac{3\pi}{5}\right)$

6. If $f(0) = f'(0) = 0$ and $f''(x) = \tan^2 x$ then $f(x)$ is

(1) $\log|\sec x| - \frac{1}{2}x^2$

(2) $\log \cos x + \frac{1}{2}x^2$

(3) $\log|\sec x| + \frac{1}{2}x^2$

(4) $x^4 + x^3 + 1$

7. $\int \frac{x^2 \tan^{-1} x^3}{1+x^6} dx =$

(1) $\frac{1}{6}(\tan^{-1} x^3)^2 + c$

(2) $-\frac{1}{6}(\tan^{-1} x^3)^2 + c$

(3) $\frac{1}{4}(\tan^{-1} x^3)^2 + c$

(4) $\frac{1}{3}(\tan^{-1} x^3)^2 + c$

8. $\int \frac{dx}{\sqrt{1-4x^2}} = k_1 f(x) + k_2$ where k_1 is constant and

k_2 is an integration constant then $2k_1 =$

- (1) 1 (2) 2
(3) 3 (4) 4

9. $\int \operatorname{cosec}^4 x dx =$

- (1) $\cot x - \frac{\cot^3 x}{3} + c$
(2) $-\cot x - \frac{\cot^3 x}{3} + c$
(3) $\cot x + \frac{\cot^3 x}{3} + c$
(4) $-\cot x + \frac{\cot^3 x}{3} + c$

10. $\int \tan^4 x dx =$

- (1) $-\frac{\tan^3 x}{3} - \tan x + x + c$
(2) $\frac{\tan^3 x}{3} - \tan x - x + c$
(3) $\frac{\tan^3 x}{3} - \tan x + x + c$
(4) $\frac{\tan^3 x}{3} + \tan x + x + c$

11. $\int \frac{\sin 2x dx}{\sqrt{9 - \sin^4 x}} =$

- (1) $\sin^{-1} \left(\frac{\sin^2 x}{3} \right) + c$
(2) $-\sin^{-1} \left(\frac{\sin^2 x}{3} \right) + c$
(3) $\sin^{-1} \left(\frac{\sin^2 x}{6} \right) + c$
(4) $\sin^{-1} \left(\frac{\sin^2 x}{4} \right) + c$

12. $\int \frac{\operatorname{cosec}(\tan^{-1} x)}{1+x^2} dx$

- (1) $\ln(1-x) + c$
(2) $\ln \left(1 + x - \frac{1}{x} \right) + c$
(3) $\ln \left(\frac{\sqrt{1+x^2}-1}{x} \right) + c$
(4) $\ln x + c$

13. $\int \frac{e^x dx}{\sqrt{5-4e^x+e^{2x}}}$

- (1) $\ln(e^x - 2 + \sqrt{5-4e^x+e^{2x}}) + c$
(2) $-\ln(e^x - 2 + \sqrt{5-4e^x+e^{2x}}) + c$
(3) $\ln(e^x + 2 + \sqrt{5-4e^x+e^{2x}}) + c$
(4) $\ln(e^x - 2 - \sqrt{5-4e^x+e^{2x}}) + c$

14. $\int \frac{dx}{1+e^x}$

- (1) $-\log|e^{(-x)}+1|+c$
(2) $\log|e^{(-x)}+1|+c$
(3) $-\log|e^{(-x)}-1|+c$
(4) $\log|e^{(x)}+1|+c$

15. $\int 10 \sin 2x \cdot \cos 2x \sqrt{\cos^2 2x - 5} dx$

- (1) $\frac{5}{3}(\cos^2 2x - 5)^{3/2} + c$
(2) $\frac{5}{3}(\cos^2 2x + 5)^{3/2} + c$
(3) $-\frac{6}{3}(\cos^2 2x - 5)^{3/2} + c$
(4) $-\frac{5}{3}(\cos^2 2x - 5)^{3/2} + c$

16. $\int \frac{(2x+1)}{(x^2+4x+1)^{3/2}} dx$

(1) $\frac{x^3}{(x^2+4x+1)^{1/2}} + C$

(2) $\frac{x}{(x^2+4x+1)^{1/2}} + C$

(3) $\frac{x^2}{(x^2+4x+1)^{1/2}} + C$

(4) $\frac{1}{(x^2+4x+1)^{1/2}} + C$

17. $\int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$

18. $\int \frac{xdx}{x^4+x^2+1}$

(1) $\frac{2}{\sqrt{3}} \tan^{-1} \left(x^2 + \frac{1}{2} \right) + C$

(2) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2-1}{\sqrt{3}} \right) + C$

(3) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2+1}{\sqrt{3}} \right) + C$

(4) $-\frac{1}{\sqrt{3}} \tan^{-1} \left(x^2 + \frac{1}{2} \right) + C$

19. $\int \frac{4x+3}{3x^2+3x+1} dx$

(1) $\frac{1}{3} \ln(3x^2+3x+1) + \frac{2}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x+\sqrt{3}) + C$

(2) $\frac{2}{3} \ln(3x^2+3x+1) + \frac{2}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x+\sqrt{3}) + C$

(3) $\frac{2}{3} \ln(3x^2+3x+1) + \frac{1}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x+\sqrt{3}) + C$

(4) $\frac{2}{3} \ln(3x^2+3x+1) - \frac{2}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x+\sqrt{3}) + C$

20. $\int \frac{dx}{2+\sin^2 x}$

(1) $\frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{\sqrt{2}} \right) + c$

(2) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{\sqrt{2}} \right) + c$

(3) $-\frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{\sqrt{2}} \right) + c$

(4) $-\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{\sqrt{2}} \right) + c$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (1)
3. (3)
4. (4)
5. (3)
6. (1)
7. (1)
8. (1)
9. (2)
10. (3)

11. (1)
12. (3)
13. (1)
14. (1)
15. (4)
16. (2)
17. $\frac{1}{4}(\sin^{-1} x)^4 + C$
18. (3)
19. (2)
20. (1)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>